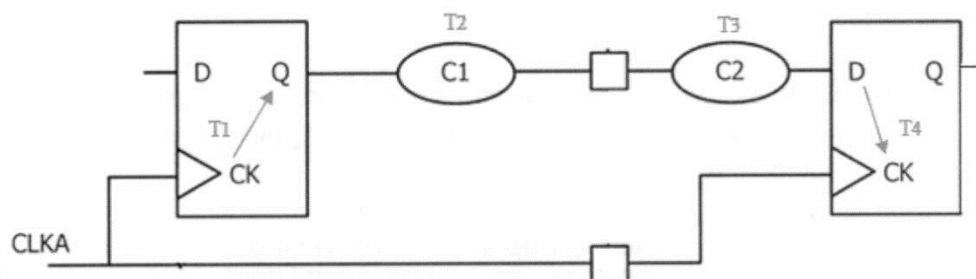


NCTU-EE IC LAB – SPRING 2016

Final Online Test - Quiz 1:30p.m. ~ 3:00p.m.

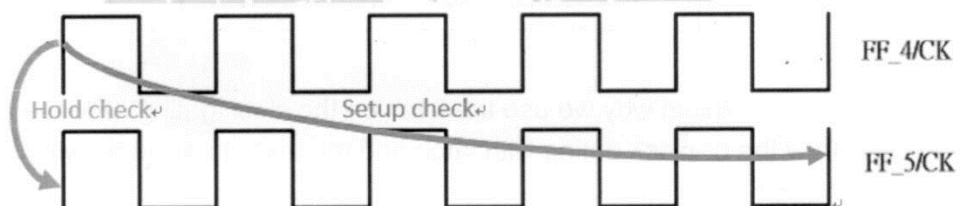
1. Please specify the following by using T1, T2, T3, T4. (12pts)

- (1) Clock cycle
- (2) Input delay
- (3) Output delay



2. Please fill in the blanks below to finish the commands for multicycle(8pts)

set_multicycle_path (?) –from FF_4/CK –to FF_5/D
set_multicycle_path (?) –hold -end –from FF_4/CK –to FF_5/D



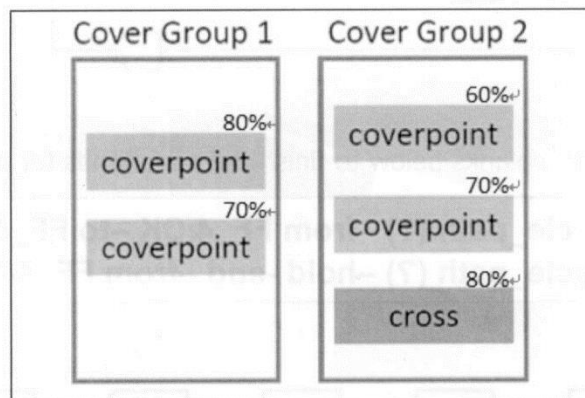
3. Please **draw the waveform** to explain the following two assertions. (10pts)

```
property p_req_ack;
  @(posedge clk) mem_en |-> (req ##2 ack);
endproperty: p_req_ack
```

```
property p_req_ack;
  @(posedge clk) mem_en |=> (req ##2 ack);
endproperty: p_req_ack
```

- f↓
pipe line
gated clock
- Please list four ways to reduce the power consumption, and briefly explain why. (12pts)
 - Please explain what is IR drop? And give two examples that how IR drop influences on design (10pts)
power R L
 - Explain antenna effect, and give two method to fix it. (10pts)
 - What is the purpose of placement blockage? How and where do we set this blockage? (10pts)
route blockage
 - Calculate the coverage of the whole design, the coverage of coverpoints or cross is specified as shown on the picture (10pts)

Design



route → placement blockage

- Please explain why we use the latch on the clock gating. (Please also draw the or-clock gating with latch and the and-clock gating with latch) (10pts)

data clock bit

- Please design a memory with following spec: (8pts)

- Number of words: 500
- Number of bits: 10

- How many input pins are needed (at least)?
- How many bits of address are needed (at least)?

10bit	Entry 1
10bit	Entry 2
10bit	Entry 3
.	.
.	.
.	.
.	.
10bit	Entry 500